

PAKS, Hungary

WMO: 129500

Lat: 46.583N

Lon: 18.850E

Elev: 322

StdP: 14.53

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.4	15.7	5.1	7.2	12.8	10.1	9.3	19.1	14.5	39.7	12.9	36.9	2.3	350	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.5	93.2	70.6	90.1	69.4	87.0	68.2	72.5	88.0	71.0	85.9	69.6	83.8	5.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.7	103.0	77.5	66.2	97.6	75.7	64.8	92.8	74.5	36.2	87.9	34.9	86.1	33.7	84.0	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.6	11.0	9.5	1.5	97.4	6.7	2.8	-3.3	99.4	-7.2	101.0	-10.9	102.6	-15.7	104.6
			0.6	75.3	6.6	1.6	-4.1	76.4	-8.0	77.3	-11.7	78.2	-16.5	79.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.1	32.1	35.2	42.9	52.9	61.3	67.7	71.1	70.5	61.3	52.3	42.8	33.7
	DBStd	15.65	8.16	9.05	7.83	6.98	6.47	6.22	5.58	5.64	6.28	7.66	7.67	8.35
	HDD50	2079	556	418	241	50	3	0	0	0	1	67	235	508
	HDD65	5301	1020	835	684	366	150	42	12	16	144	395	665	972
	CDD50	2836	0	3	23	135	353	532	654	635	341	139	20	1
	CDD65	584	0	0	0	2	35	123	202	186	34	2	0	0
	CDH74	7563	0	0	0	93	608	1583	2468	2273	487	51	0	0
	CDH80	3015	0	0	0	15	159	605	1094	1006	135	1	0	0
Wind	WSAvg	4.0	4.0	4.4	4.9	4.6	4.2	3.9	4.0	3.4	3.4	3.4	3.8	4.0
Precipitation	PrecAvg	23.5	1.3	1.4	1.2	1.5	2.3	2.6	2.6	2.2	2.4	2.1	2.0	1.8
	PrecMax	37.8	3.2	3.4	4.9	3.8	7.6	4.9	6.1	6.7	6.6	4.8	5.3	4.3
	PrecMin	15.6	0.2	0.0	0.2	0.0	0.5	0.5	1.0	0.2	0.4	0.1	0.0	0.0
	PrecStd	5.1	0.8	1.0	0.9	0.9	1.4	1.4	1.5	1.4	1.6	1.3	1.1	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.0	62.1	71.1	82.2	88.3	94.1	97.9	97.8	88.9	79.0	67.0	56.0
		MCWB	48.6	50.1	53.7	61.0	65.9	70.7	70.9	70.3	67.3	63.8	57.0	50.4
	2%	DB	50.8	57.3	66.3	76.3	84.3	90.5	92.9	93.4	83.9	74.9	62.0	50.7
		MCWB	45.6	47.9	52.0	57.9	64.9	70.1	70.8	70.4	65.6	61.5	54.7	46.5
	5%	DB	46.9	52.7	61.7	72.5	81.0	86.9	89.8	89.6	79.6	71.0	58.2	47.1
		MCWB	42.8	45.7	49.8	56.2	63.4	68.8	69.6	69.1	63.9	59.6	52.5	43.9
	10%	DB	43.5	48.4	57.4	68.5	76.9	83.3	86.6	85.8	75.6	66.4	54.8	44.5
		MCWB	40.5	43.3	47.8	54.4	61.8	67.0	68.5	67.5	62.3	57.5	50.2	41.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.3	51.1	55.7	62.6	68.3	73.8	74.5	74.2	69.1	65.1	58.2	50.9
		MCDB	54.6	59.8	67.0	78.6	83.8	88.9	89.7	89.6	83.4	76.4	64.9	55.2
	2%	WB	46.1	48.7	53.0	59.6	66.1	71.7	72.4	72.0	66.9	62.5	55.4	47.1
		MCDB	50.2	56.0	63.8	72.9	80.7	86.4	88.3	87.5	80.1	72.1	60.8	50.0
	5%	WB	43.3	46.2	51.0	57.5	64.5	69.9	70.9	70.6	65.3	60.4	52.9	44.5
		MCDB	46.4	51.3	60.2	69.3	78.1	83.8	86.0	85.6	76.8	69.1	57.8	46.7
	10%	WB	40.6	44.0	48.6	55.5	62.7	68.2	69.5	69.0	63.5	58.3	50.6	42.2
		MCDB	43.1	48.2	56.3	66.4	75.0	80.7	83.5	83.1	73.1	65.5	54.3	44.2
Mean Daily Temperature Range	5% DB	MDBR	12.3	16.0	20.2	23.3	23.7	23.7	24.5	25.3	22.5	21.8	14.9	11.2
		MCDBR	17.9	23.4	28.6	31.5	31.3	30.1	31.0	32.3	30.0	28.0	21.0	15.4
		MCWBR	13.1	15.6	16.6	15.8	13.9	12.1	11.4	11.9	13.6	15.7	14.2	11.8
	5% WB	MCDBR	17.4	20.6	25.1	27.5	28.1	27.7	27.6	28.4	26.3	25.5	19.6	14.6
		MCWBR	13.2	14.3	15.0	14.3	13.0	11.6	11.0	11.4	12.5	14.8	13.6	11.5
Clear-Sky Solar Irradiance	taub	0.336	0.359	0.395	0.447	0.434	0.433	0.444	0.446	0.425	0.390	0.361	0.335	
	taud	2.365	2.297	2.229	2.094	2.196	2.232	2.202	2.206	2.229	2.315	2.348	2.369	
	Ebn at Noon	236	253	260	256	265	265	260	254	247	238	222	221	
	Edh at Noon	24	31	38	48	44	43	44	42	38	30	24	21	
All-Sky Solar Radiation	RadAvg	357	617	999	1462	1760	1914	1899	1705	1197	791	439	274	
	RadStd	45	93	136	137	168	142	142	168	138	82	40	41	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.92	N/A	N/A	N/A	+1.55	+1.19	N/A	N/A	N/A	N/A
Regional (47 neighbors)		+1.12	N/A	N/A	N/A	+0.90	+0.97	-211	-310	+176	+99

Nomenclature: See separate page